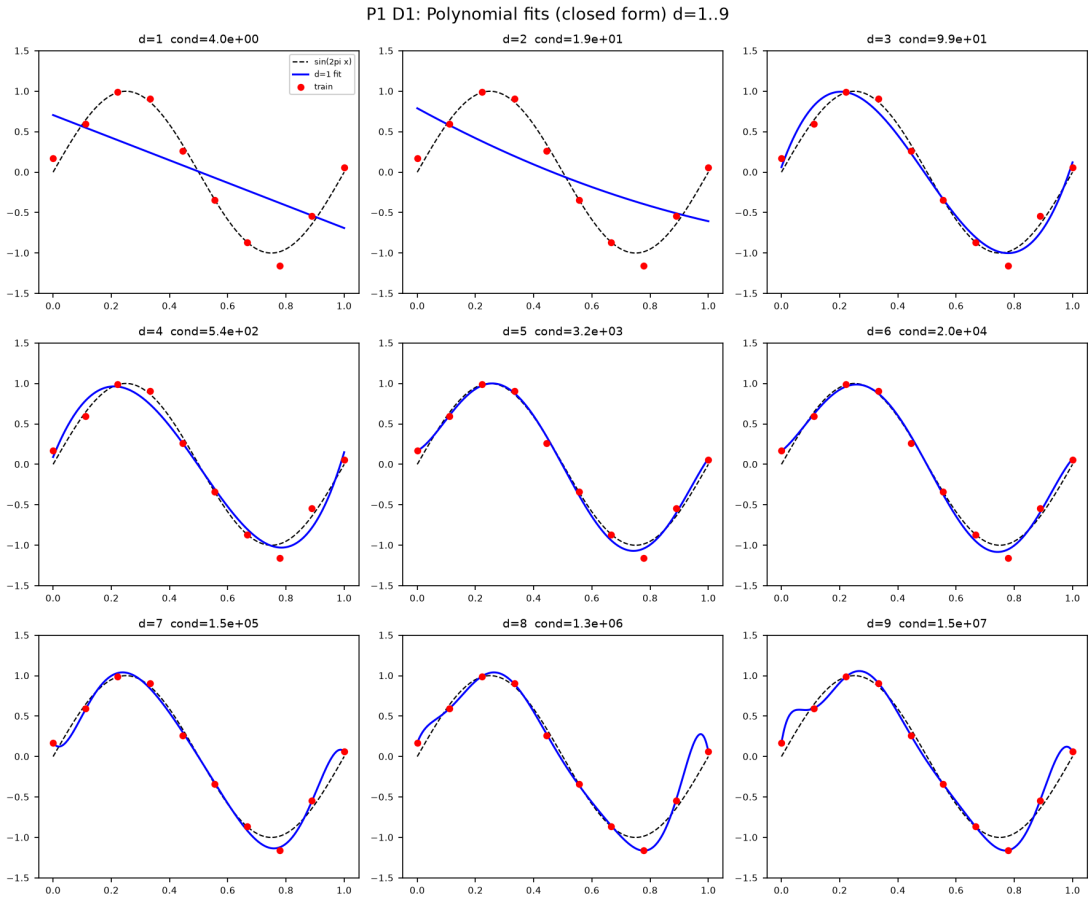
CS F320 Foundations of Data Science - Assignment 1

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Setup: Python with NumPy implemented from scratch (no scikit-learn); random seed 42 for SGD and mini-batch shuffling.

P1 – Closed form and the gradient-descent family

D1 – Closed-form fits and condition numbers



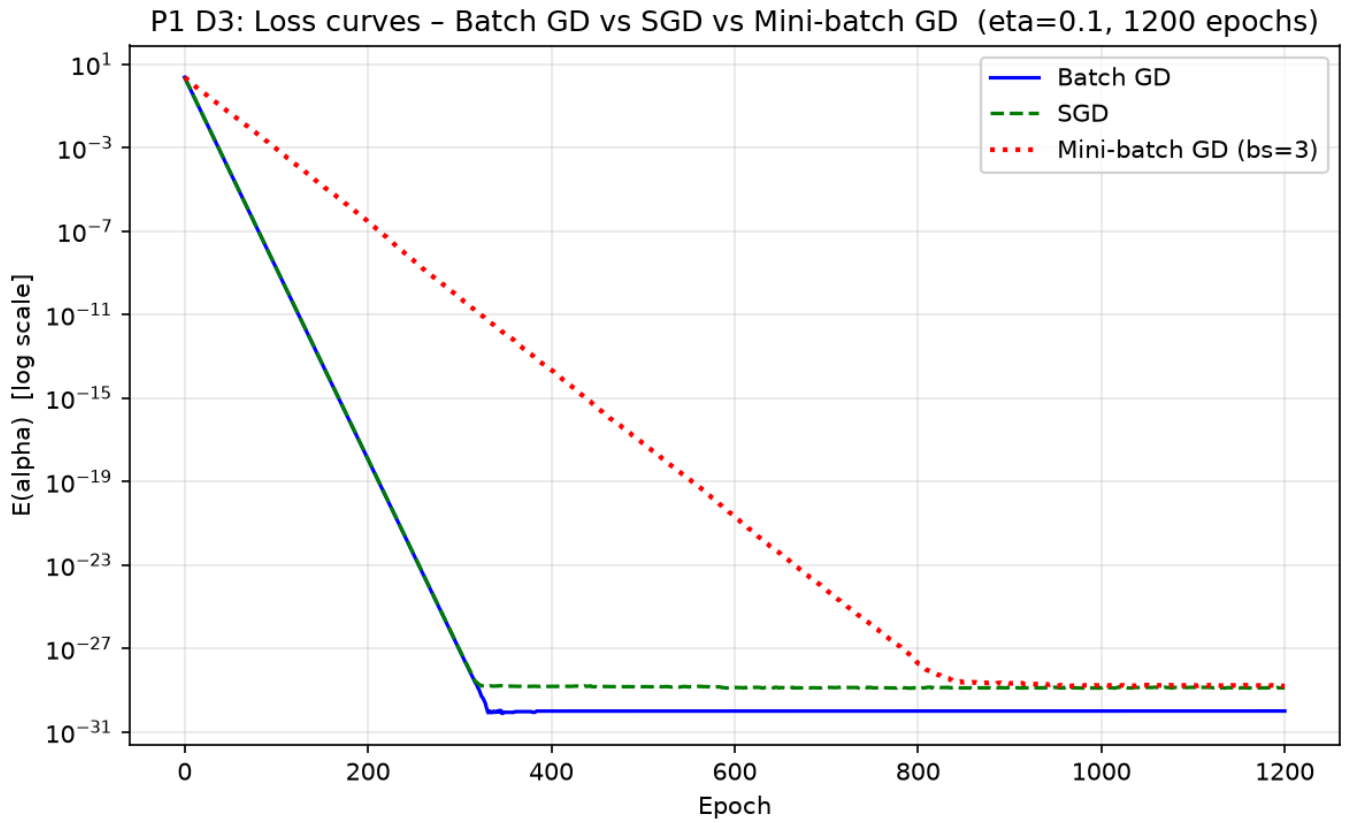
Closed-form polynomial fits, degrees 1–9

Degree	cond(Phi)
1	3.9849e+00
2	1.9204e+01
3	9.9026e+01
4	5.4143e+02
5	3.1696e+03
6	2.0266e+04
7	1.4628e+05
8	1.2660e+06
9	1.5193e+07

D2 – QR reparametrisation (d = 9)

```
Q shape: (10, 10), R shape: (10, 10)
Max off-diagonal of Q Q^T: 7.7716e-16
```

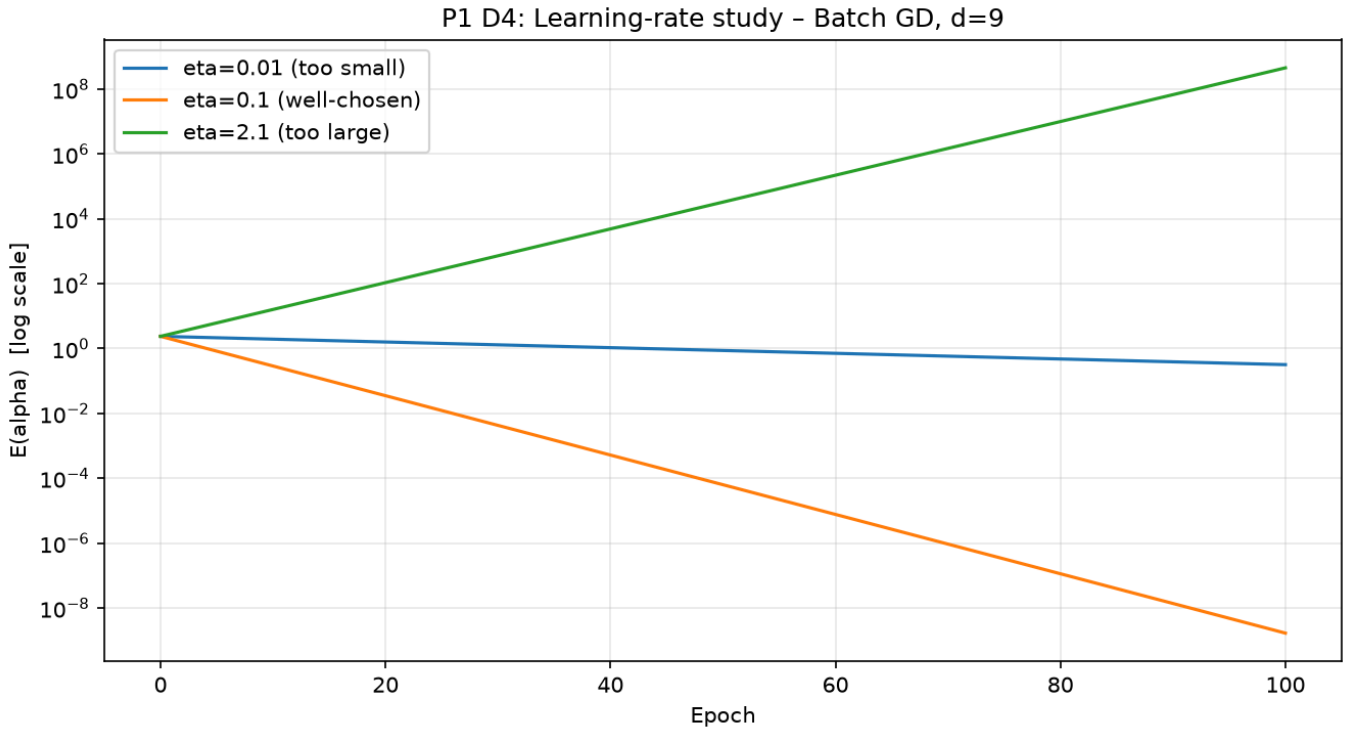
D3 – Loss curves



Batch GD, SGD and mini-batch GD loss (eta = 0.1)

Method	Final E
Batch GD	1.006972e-30
SGD	1.291692e-29
Mini-batch GD	1.562047e-29

D4 – Learning-rate study



Learning-rate study, 100 epochs

eta	Final E (epoch 100)
0.01	3.189640e-01
0.1	1.679595e-09
2.1	4.521056e+08
2 (special case)	2.380690e+00

```
eta=0.01: w = [1.0717e-01 1.3484e+01 -2.6217e+02 2.4015e+03 -1.1151e+04 2.8913e+04
 -4.4003e+04 3.9059e+04 -1.8683e+04 3.7124e+03]
eta=0.1: w = [1.6905e-01 2.1268e+01 -4.1352e+02 3.7879e+03 -1.7588e+04 4.5604e+04
 -6.9408e+04 6.1608e+04 -2.9468e+04 5.8557e+03]
eta=2.1: w = [-2.3295e+03 -2.9307e+05 5.6983e+06 -5.2197e+07 2.4236e+08 -6.2843e+08
 9.5644e+08 -8.4896e+08 4.0607e+08 -8.0691e+07]
```

D5 – Verification against closed form (2000 epochs)

Method	||w_GD - w_cf||_2	Relative error
Batch GD	3.178392e-06	2.911191e-11
SGD	3.178918e-06	2.911674e-11
Mini-batch GD	3.178157e-06	2.910976e-11

```
||w_closed||_2 = 1.091784e+05
cond(R) = 1.5193e+07
```

D6 – Discussion

With the QR reparametrisation Q is orthogonal, so E(alpha) has curvature 1 in every direction and each full pass over the data contracts the error by |1 - eta| = 0.9 (eta = 0.1); the loss curves in the figure are straight lines on the log scale for that reason. Batch GD and SGD lie on top of each other: the rows of Q are

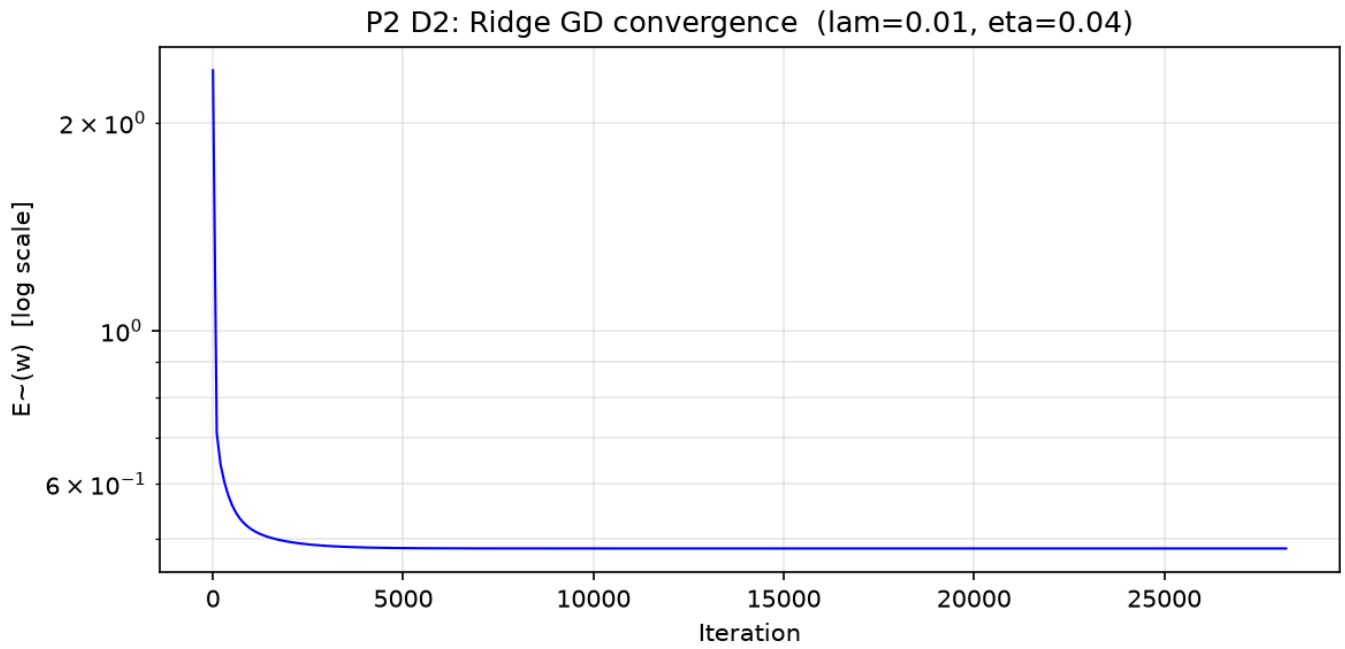
orthonormal, so each single-sample step in SGD touches one alpha-direction and one epoch updates every direction once, exactly as batch GD does. Mini-batch GD descends visibly more slowly because averaging the gradient over a batch shrinks the per-sample step, yet all three reach a round-off floor (final E 1.006972e-30, 1.291692e-29 and 1.562047e-29 respectively).

P2 – Ridge regression as MAP

D1 – Closed-form ridge

Ridge weights are obtained with np.linalg.solve on (Phi^T Phi + lam I) w = Phi^T t; the resulting fits and errors are reported in D3 and D4.

D2 – Gradient-descent agreement and conditioning

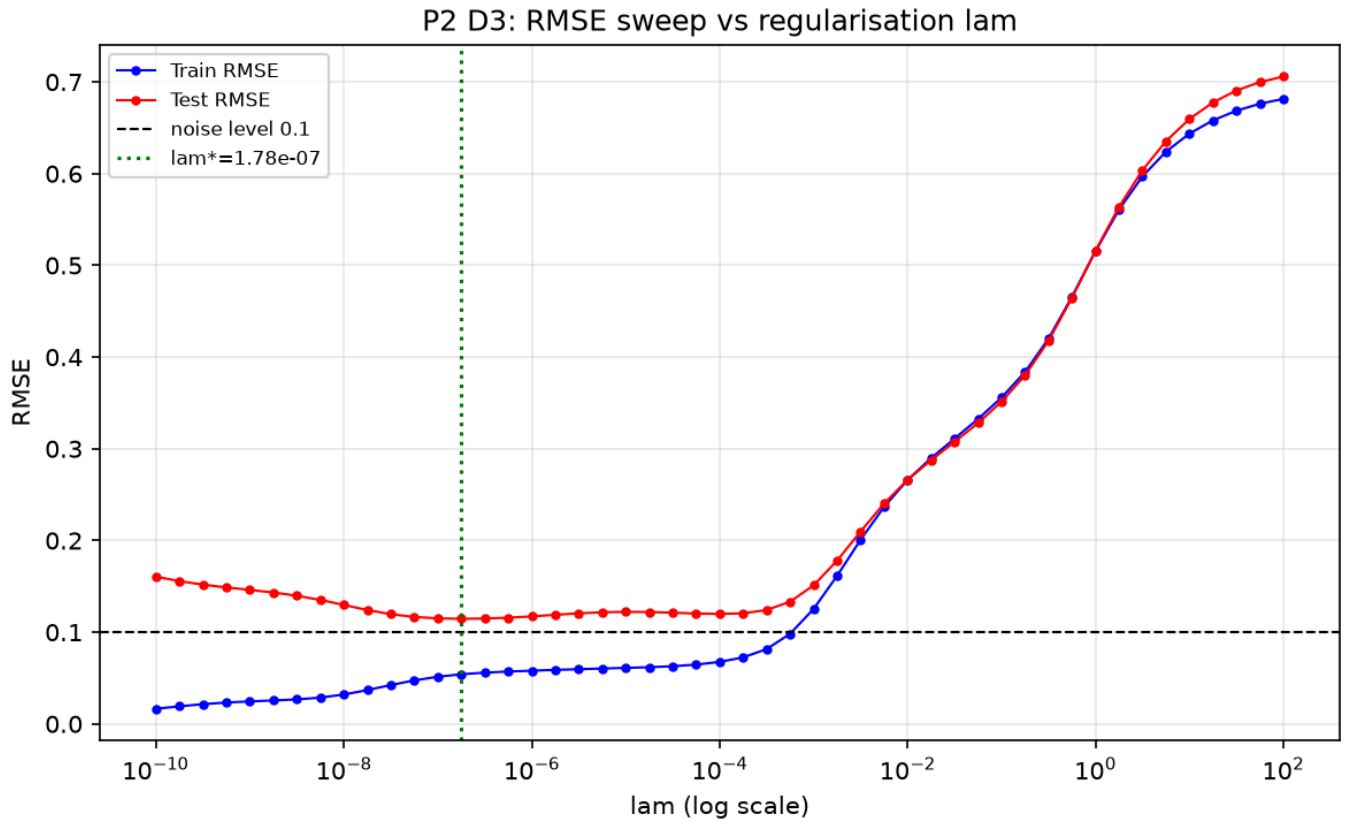


Ridge GD loss vs iteration (lam = 0.01)

Quantity	Value
cond(Phi^T Phi)	2.3124e+14
cond(Phi^T Phi + lam I), lam = 0.01	2.0381e+03
GD iterations to converge	28227
Relative error vs closed form	9.9968e-07

Without the penalty the normal matrix is extremely ill-conditioned (cond = 2.3124e+14), whereas adding lam = 0.01 on the diagonal brings it down to 2.0381e+03, which is why gradient descent converges (28227 iterations, relative error 9.9968e-07).

D3 – RMSE sweep and the lam interval

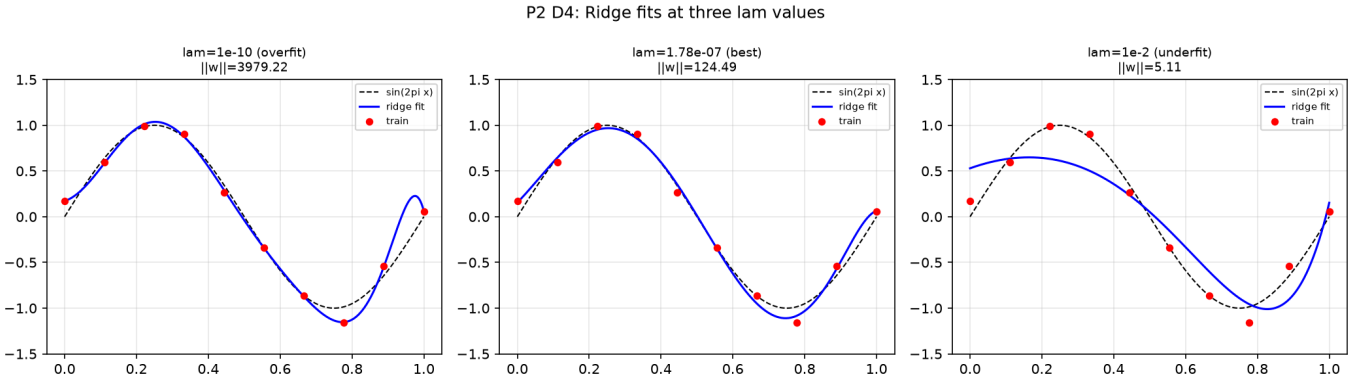


Train and test RMSE vs lam

Quantity	Value
lam*	1.7783e-07
Test RMSE at lam*	0.114577
Interval (test RMSE within 5% of min)	[3.1623e-08, 1.0000e-04]

The test-RMSE minimum is flat rather than sharp: every lam in [3.1623e-08, 1.0000e-04] stays within 5% of the minimum (0.114577 at lam* = 1.7783e-07), so lam need not be tuned precisely. Below 3.1623e-08 the model overfits (train RMSE keeps falling while test RMSE rises), and above 1.0000e-04 it underfits (both RMSEs rise as the weights are shrunk too hard).

D4 – Fitted curves

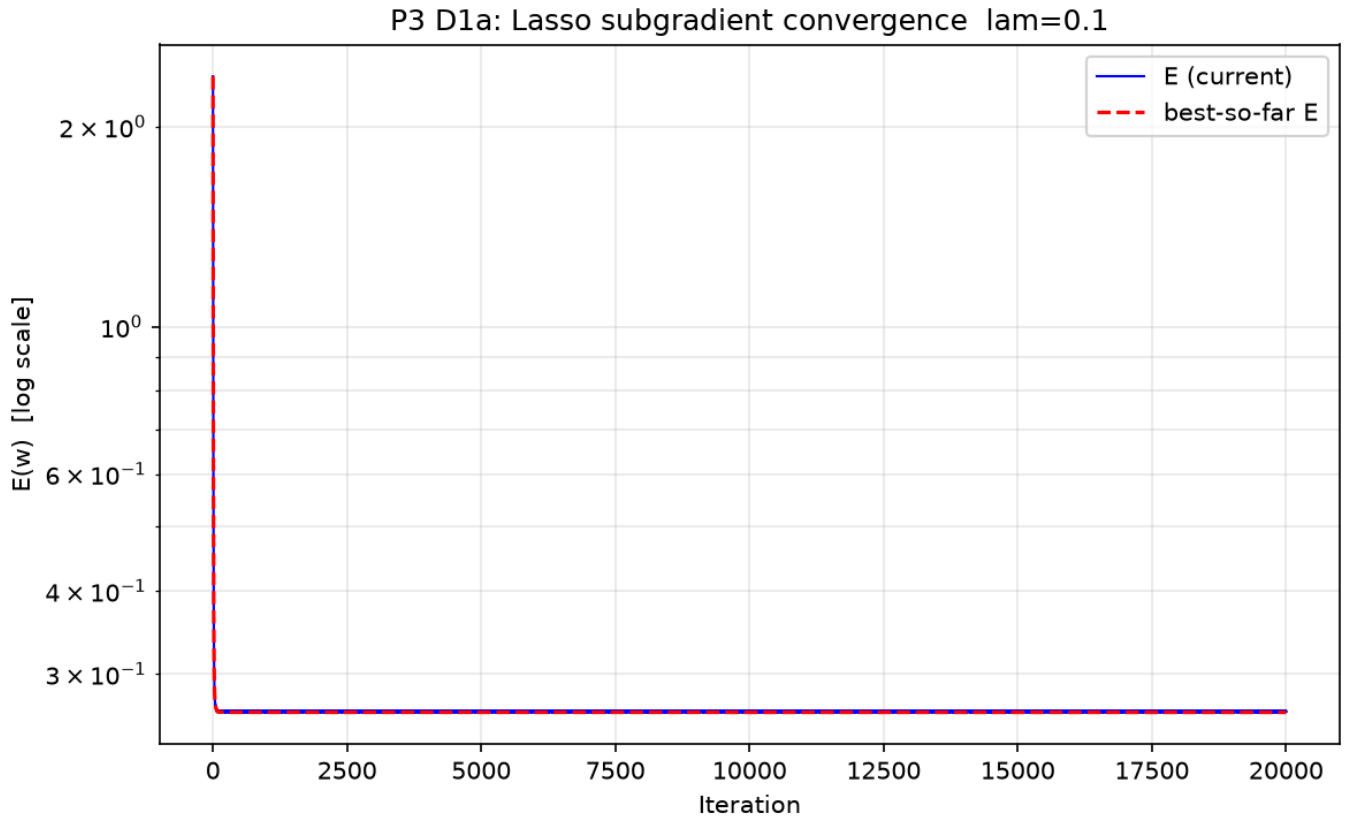


Fitted curves at lam = 1e-10, lam* and lam = 1e-2

lam	Regime	||w||_2
lam = 1e-10	overfit	3979.2246
lam = 1.78e-07	best	124.4946
lam = 1e-2	underfit	5.1079

P3 – Lasso via subgradient descent

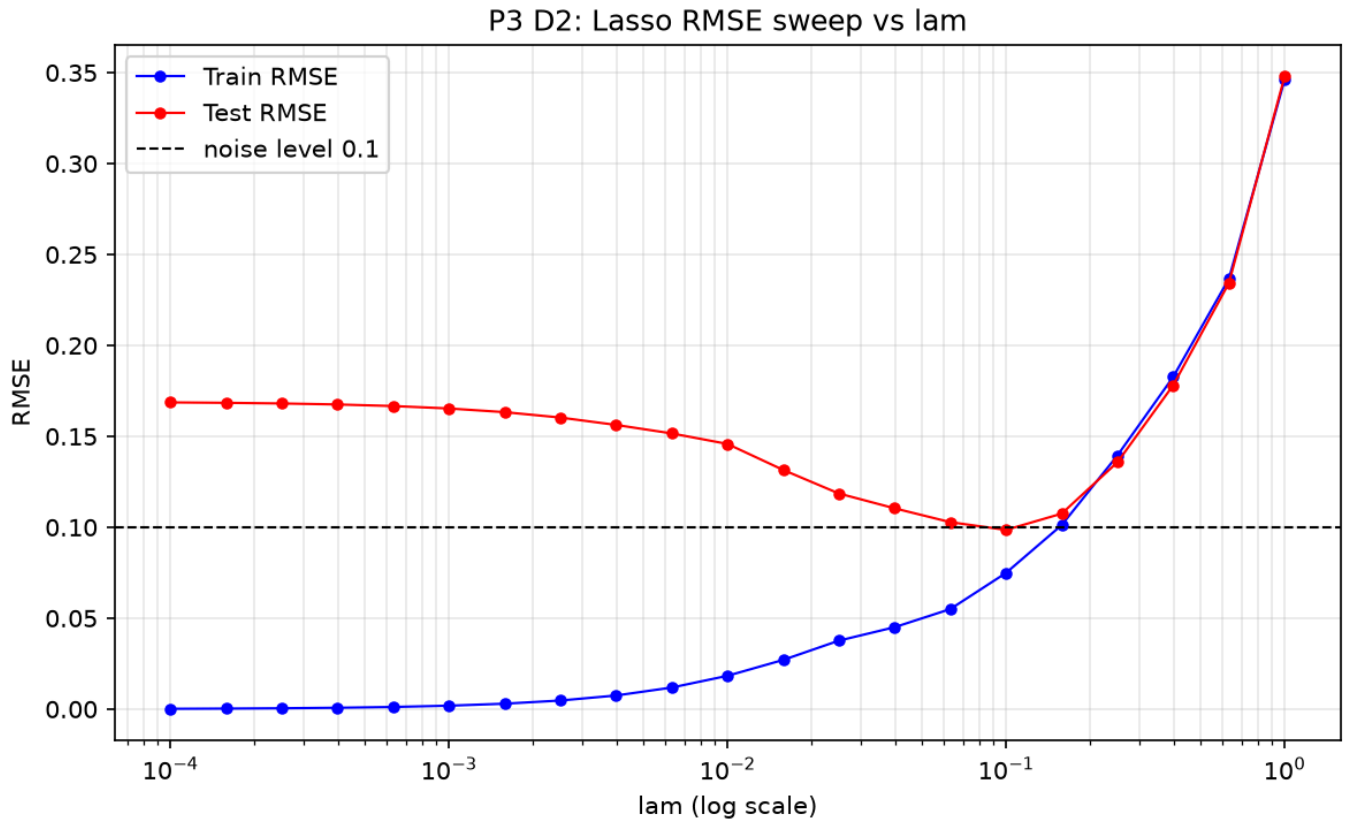
D1 – Detailed analysis at lam = 0.1



Subgradient loss and best-so-far at lam = 0.1

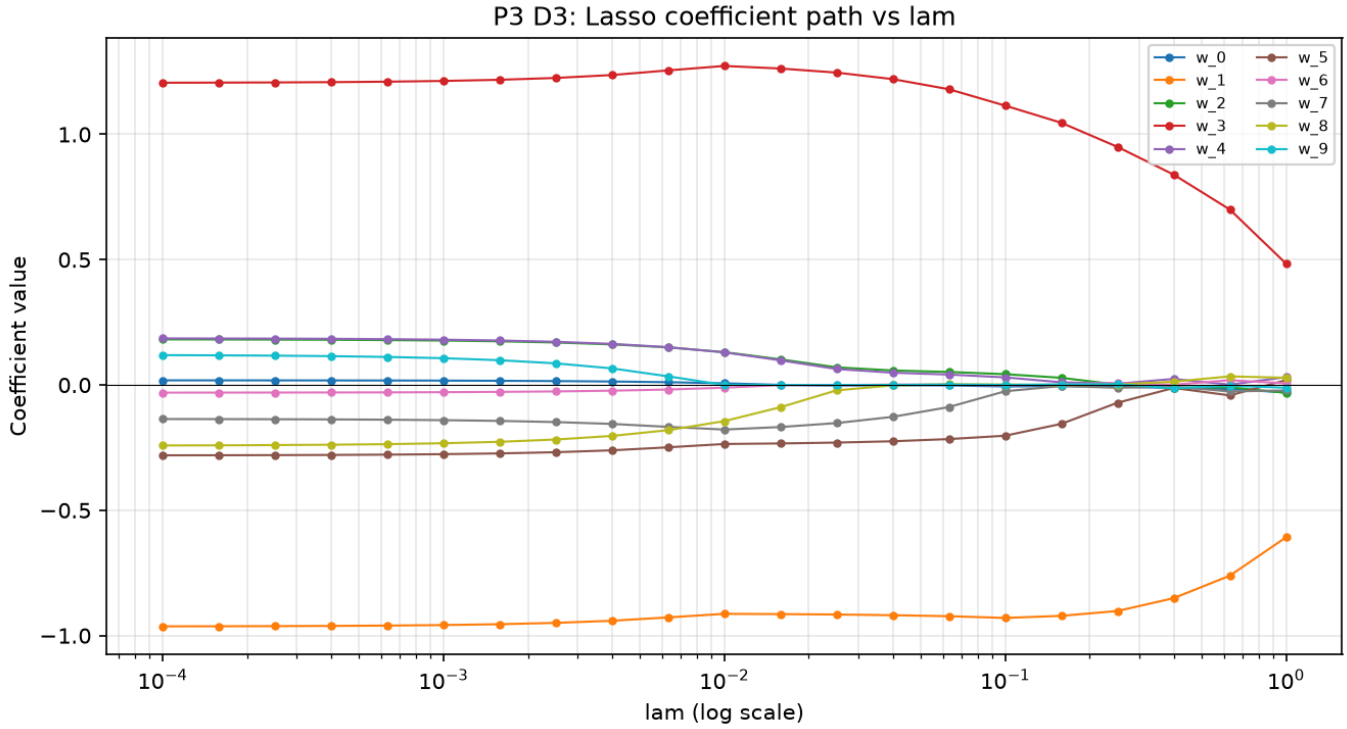
Quantity	Value
Final E	0.263570
Best E	0.262619
E - E*	0.001070
E* (3-decimal reference)	0.2625
||w - w*||_2	0.007636
max |Phi_j^T r - lam sign(w_j)| (non-zero w_j)	1.435812e-02
max |Phi_j^T r| / lam (zero w_j)	0.683041
Non-zero coefficient indices	[1, 2, 3, 4, 5, 7]
Zero coefficient indices	[0, 6, 8, 9]

D2 – RMSE sweep



Lasso train and test RMSE vs lam

D3 – Coefficient path



Coefficient path vs lam

lam	# ~zero coefficients (|w_j| < 2*eta*lam)
1.0000e-04	0
1.5849e-04	0
2.5119e-04	0
3.9811e-04	0
6.3096e-04	0
1.0000e-03	0
1.5849e-03	0
2.5119e-03	0
3.9811e-03	0
6.3096e-03	0
1.0000e-02	1
1.5849e-02	3
2.5119e-02	3
3.9811e-02	4
6.3096e-02	4
1.0000e-01	4
1.5849e-01	6
2.5119e-01	7
3.9811e-01	8
6.3096e-01	8
1.0000e+00	8

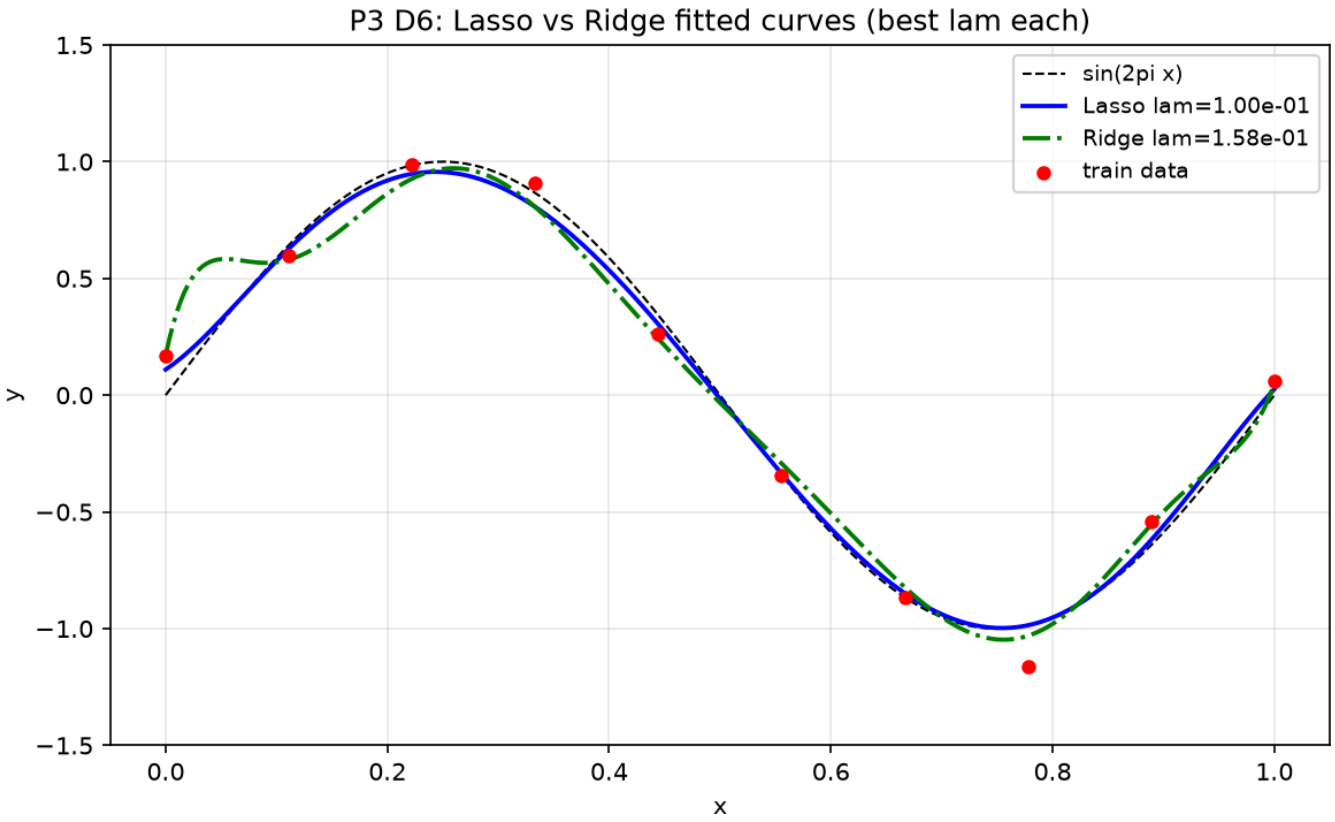
D4 – Best lam

Quantity	Value
lam*	1.0000e-01
Test RMSE at lam*	0.098544
# ~zero coefficients	4
Indices	[0, 6, 8, 9]

D5 – Lasso vs ridge (Legendre basis)

Method	lam*	Test RMSE	||w||_2	# coeffs |w_j| < 0.01
Lasso	1.0000e-01	0.098544	1.4667	4
Ridge	1.5849e-01	0.142233	1.4406	1

D6 – Fitted curves



Lasso vs ridge at the best lam of each

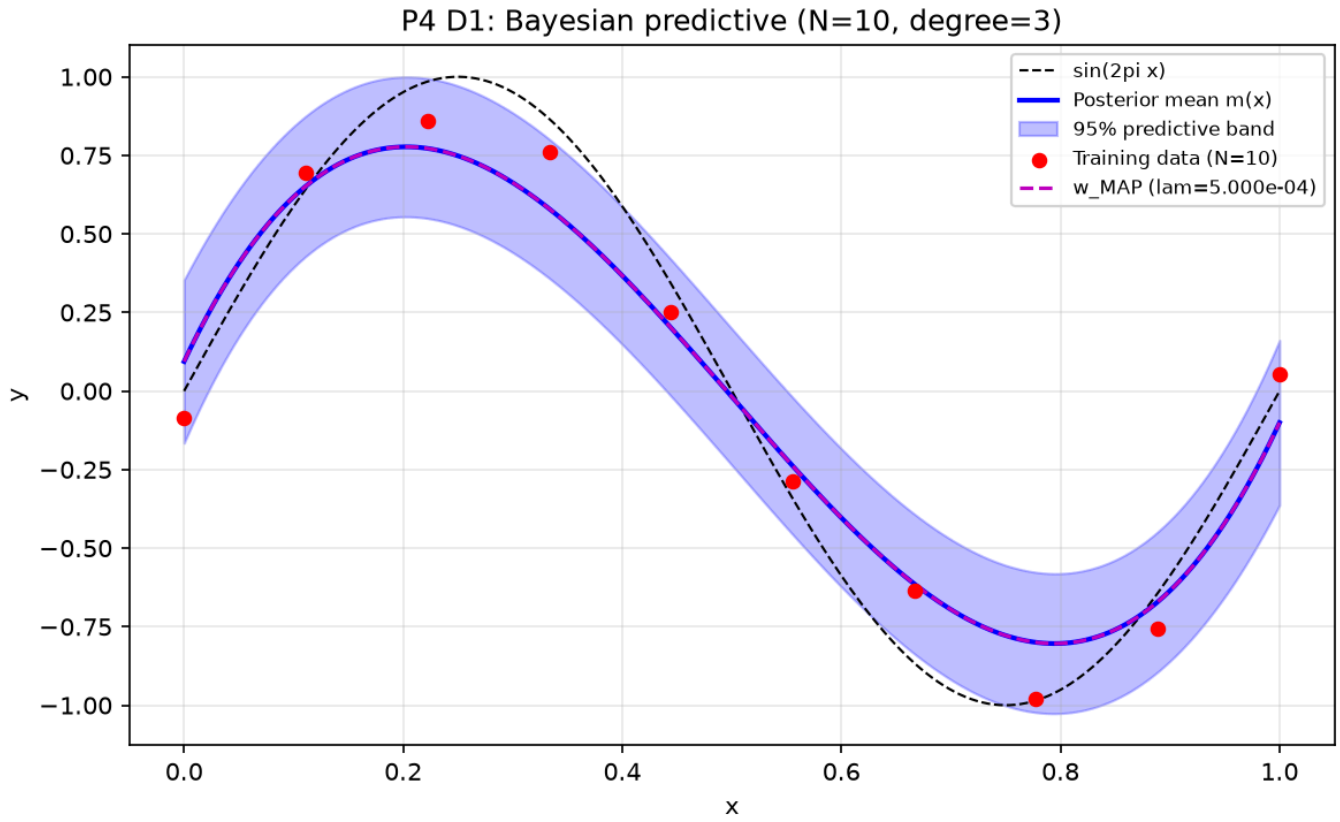
D7 – Discussion

The L1 penalty yields exact zeros because a zero coefficient is optimal whenever the subgradient condition |Phi_j^T r| <= lam holds (here max |Phi_j^T r|/lam = 0.683041 on the zero set), and the L1 ball has corners on the axes; the L2 penalty only shrinks coefficients, which is why ridge leaves 1 coefficient(s) below 0.01 against 4 for lasso. For lam ≥ lam* = 1.0000e-01 the train RMSE and the test RMSE both increase as lam grows, because the extra sparsity removes model capacity that the data support and the fit moves from the noise level (0.098544 at lam*) towards underfitting.

P4 – Bayesian linear regression

D1 – Posterior predictive (N = 10)

Hyperparameters: alpha=0.05, beta=100.0, degree=3.

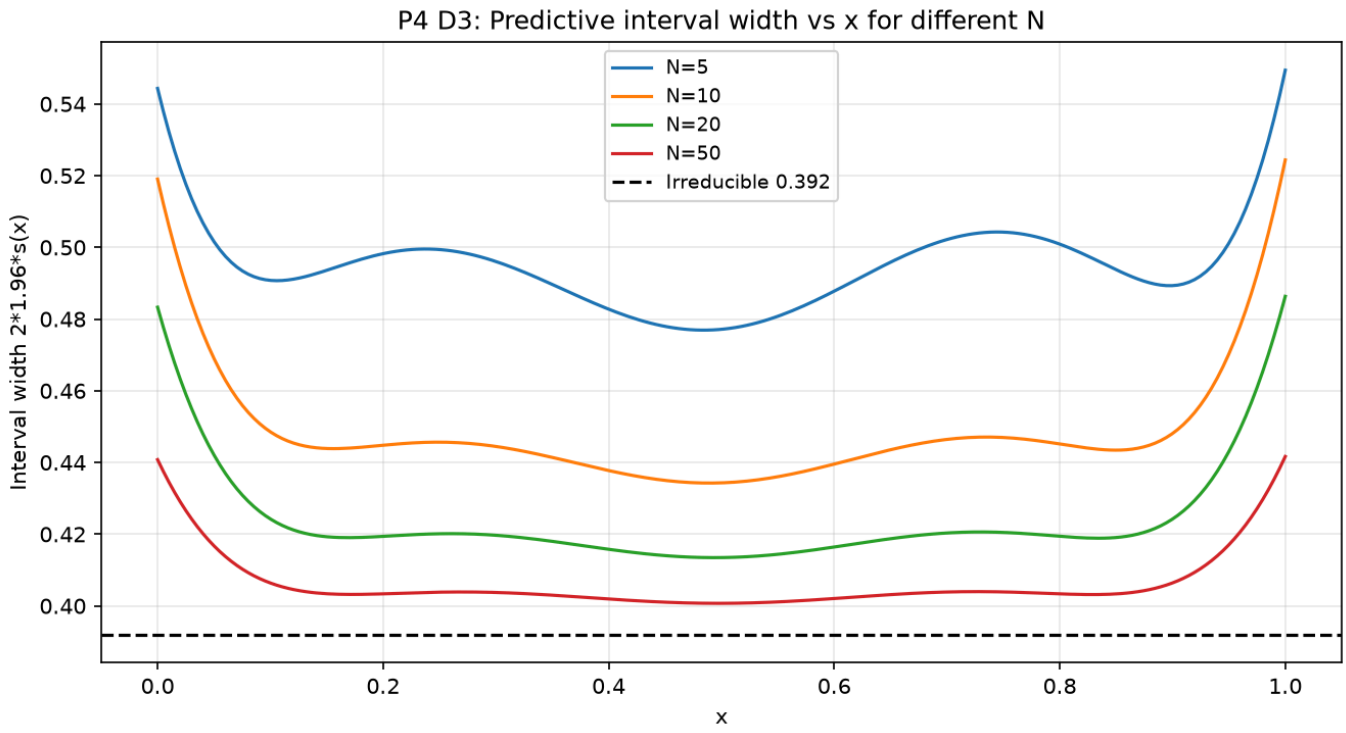


Posterior mean, 95% band and w_MAP overlay (N = 10)

D2 – Equivalence of posterior mean and MAP

```
lam = alpha/beta = 5.000000e-04
||w_MAP - m_N||_2 = 1.1276e-12
```

D3 – Interval width vs N



Predictive interval width 2*1.96*sigma(x) for N = 5, 10, 20, 50

N	Mean width	Max width
5	0.4952	0.5494
10	0.4491	0.5244
20	0.4242	0.4863
50	0.4064	0.4417

Irreducible floor 2*1.96/sqrt(beta) = 0.3920.

D4 – Discussion

As N grows the data term beta*Phi^T Phi dominates the posterior precision, so S_N shrinks and the parameter-uncertainty part phi^T S_N phi of the predictive variance falls; the mean interval width drops from 0.4952 at N=5 to 0.4064 at N=50. It cannot go below the floor 2*1.96/sqrt(beta) = 0.3920, because the noise term 1/beta in the predictive variance is irreducible however much data we collect.

Notes

- P1 solves the polynomial fit in the QR reparametrisation (Phi = QR, alpha = R w), where Q^T Q = I and every curvature equals 1; weights are recovered as w = R^-1 alpha with a linear solve.
- No explicit matrix inverses are used for the weights: np.linalg.lstsq (P1 closed form) and np.linalg.solve (P2, P4). Only the P4 posterior covariance, needed for the predictive variance, uses np.linalg.inv.
- P3 treats a coefficient as ~0 when |w_j| < 2*eta*lam, the chatter amplitude of constant-step subgradient descent; the P3 D5 comparison and the D1 zero/non-zero split use the fixed rule |w_j| < 0.01.
- Subgradient descent with a constant step does not converge exactly: iterates chatter around the optimum with amplitude of order eta*lam, so the final E and ||w - w*|| keep a small non-zero gap (P3 D1), and the best-so-far E is also reported.
- The P3 reference E* and w* are 3-decimal values from the assignment, so gaps below about 1e-3 are within their rounding.